



Core Video Resources: Regime-Shift Asset Allocation

Welcome to the quantitative finance deep-end! The concepts in this project—Hidden Markov Models, Convex Optimization, and Walk-Forward Validation—are incredibly powerful, but they can be heavy to learn just from reading text.

We have curated a list of the best YouTube videos to help you visualize the math and see the Python implementation in action. Watch these alongside the Project Guide.ipynb notebook.

1. Demystifying Hidden Markov Models (HMMs)

Your goal is to detect the "hidden" state of the market (e.g., Bull vs. Bear) using visible data (prices and volatility). These videos explain the intuition behind the `hmmlearn` library.

- [Hidden Markov Model: Data Science Concepts](#) by *ritvikmath* (13 mins)
 - **Why watch it:** This is the absolute best whiteboard explanation of what an HMM actually is. If you are confused by transition matrices and hidden states, start here. It strips away the complex math and focuses on intuition.
- [Hidden Markov Models for Quant Finance](#) by *Roman Paolucci* (56 mins)
 - **Why watch it:** This is a deep dive specifically tailored for quantitative finance. It shows how quants use HMMs to predict market regimes using historical data, which is exactly what you are doing in Phase 2 of this project.

2. Convex Portfolio Optimization (CVXPY)

Once you know the market regime, how much money goes into stocks vs. bonds? You will use CVXPY to mathematically optimize the portfolio.

- [Convex Optimization in Python with CVXPY](#) by *Steven Diamond @ SciPy 2018* (29 mins)
 - **Why watch it:** Steven Diamond is the co-creator of the CVXPY library! This is an excellent, beginner-friendly introduction to setting up objective functions and constraints in Python.
- [Portfolio Optimization in Python: Boost Your Financial Performance](#) by *Ryan O'Connell, CFA, FRM* (23 mins)
 - **Why watch it:** A highly practical walkthrough of setting up a portfolio, downloading data (like `yfinance`), and optimizing weights based on risk and return. It bridges the gap between pure math and actual financial data.

3. The Deadliest Trap: Lookahead Bias & Backtesting

The problem statement mentions that avoiding Lookahead Bias is "the hard and genuinely interesting part" of this project. It is the #1 reason why quant strategies make millions on paper but lose money in real life.

- [Look-Ahead Bias - Future Data | Financial Data Science](#) by *Quantra / Dr. Ernest P. Chan* (3 mins)

- **Why watch it:** A quick, punchy explanation from a veteran quant (Ernie Chan) on how future data accidentally leaks into past training sets.
- [**3 Backtesting Pitfalls That Ruin Your Trading Strategy**](#) by *Roman Paolucci* (14 mins)
 - **Why watch it:** An excellent breakdown of the major pitfalls when testing a strategy in Python. Pay close attention to how he explains data leakage, as you will need to actively avoid this during your Walk-Forward validation (Phase 4).



Pro-Tip for the Project

Don't try to watch all of these at once.

1. Watch the **ritvikmath** video when you start the hmmlearn section.
2. Watch the **CVXPY** tutorials when you get to the optimization phase.
3. Watch the **Lookahead Bias** videos right before you start building your backtesting loop.

Good luck building your engine!